

The Diagonal: a Worked Example

How a read finds its place in the genome

Every match between a read and the reference is called an anchor: a pair (*position in the read, position in the reference*). Its diagonal is simply

$$\text{diagonal} = (\text{position in reference}) - (\text{position in read}).$$

The idea is intuitive: when a read really comes from one spot in the genome, each true match advances the read *and* the reference by the same amount, so all its anchors land on one shared diagonal. Errors and repeats match random places, so they scatter across many different diagonals.

Setup. We align the read GATTACA (positions 0 to 6), which truly comes from reference position 10, and we cut it into 3-mers:

Read = G A T T A C A

Reference = C A T G A T C C A A G A T T A C A G G T A C A T T (pos. 0 to 24)

The read's five 3-mers are GAT, ATT, TTA, TAC, ACA. We look up where each one occurs in the reference; every hit is an anchor, and we compute its diagonal.

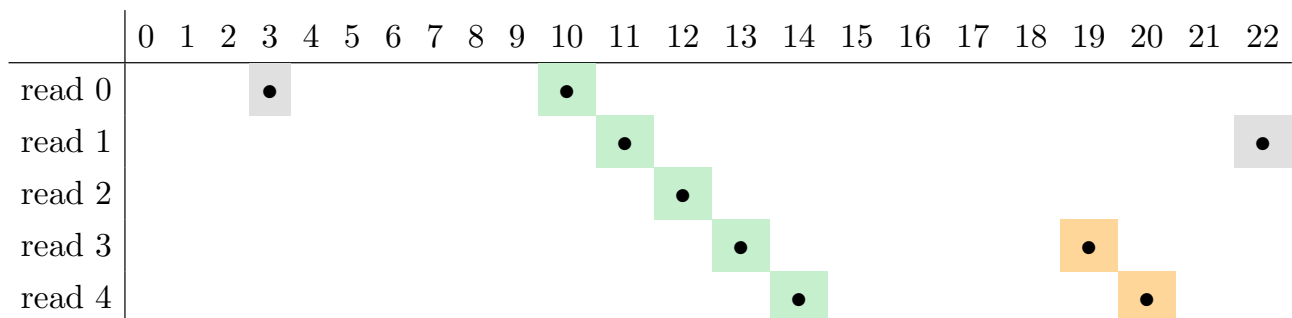
All anchors (k-mer hits)				
k-mer	read	ref	diag.	kind
GAT	0	3	3	noise
GAT	0	10	10	true
ATT	1	11	10	true
ATT	1	22	21	noise
TTA	2	12	10	true
TAC	3	13	10	true
TAC	3	19	16	repeat
ACA	4	14	10	true
ACA	4	20	16	repeat

Group by diagonal = "support"

diagonal	anchors	meaning
10	5	the alignment
16	2	repeat run
3	1	noise
21	1	noise

A real alignment piles *many* anchors onto *one* diagonal; junk scatters as lonely hits (support = 1).

The picture (a dot-plot). Rows are read positions, columns are reference positions; each • is a match. The true alignment is the long green line stepping down-right (all on diagonal 10). The repeat makes a short orange segment (diagonal 16); the grey hits are isolated noise.



Why it matters. The alignment stage only wants that green diagonal. Since every k-mer anchor has the same length, ranking anchors by length is a coin flip and can throw the true run away, keeping grey noise instead. Ranking by diagonal support (how many anchors share a diagonal) keeps the crowded green diagonal and drops the lonely noise, so the real alignment always survives.